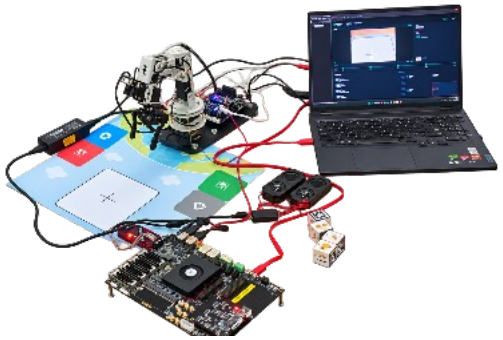
Xinyu SmartSort

Zynq-based voice and vision co-design for intelligent waste sorting

AMD Zynq UltraScale+ heterogeneous platform



INTRODUCTION

Hears · Sees · Sorts Right

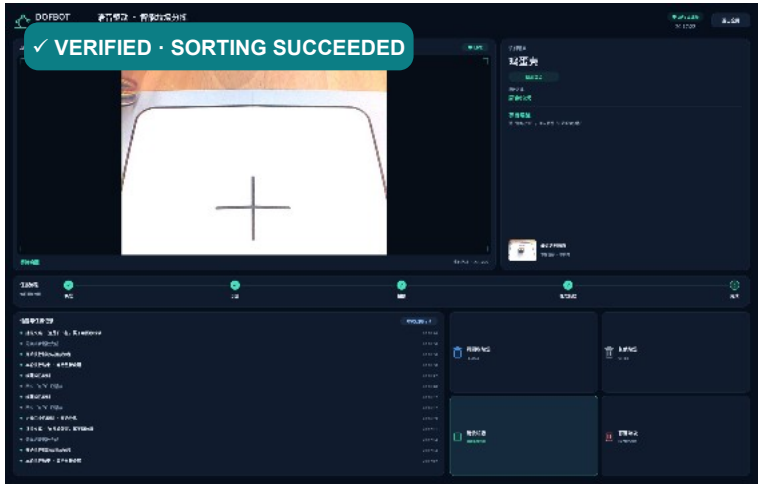
Say the word and the arm sorts — voice, vision and disposal become one interaction that makes green living visible and co-design teachable.

Wake-up — **Vision** — **Announce** — **Pick**

“Hello Xiaoya” → “Start sorting”

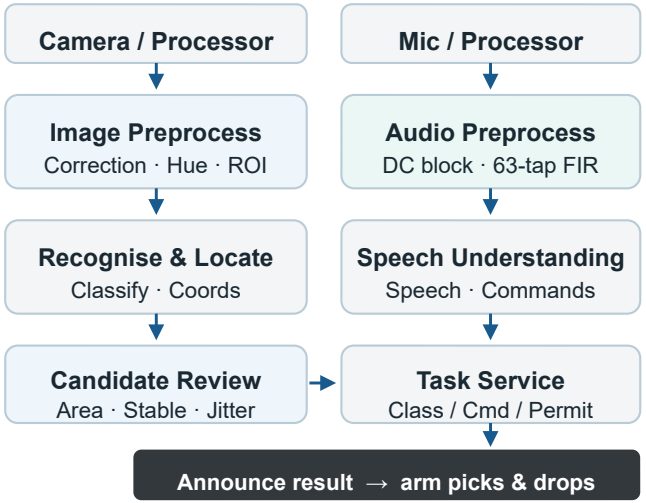
Sorting as a hands-on classroom

From “what is this?” to “where does it go?”, visitors watch the recognition, the announcement and the drop — built for eco-classes and robotics courses.



Demo: one wake-up per task, announce then pick — sorting succeeded

A small tabletop linking green living to engineering practice



Colour side = programmable logic, grey = processor

Image and sound, handled together

1. Image pipeline: colour correction, hue mask and region statistics, frame after frame.
2. Audio: 16 kHz input, DC block and a 63-tap FIR before command recognition.
3. Split of work: the processor recognises, the programmable logic moves data.

One more gate before the pick

1. Re-check area, aspect, continuity and jitter.
2. One wake-up per task: announce, then pick.
3. Busy commands never pile up; every step stays visible.

CREATIVE DESIGN

TECHNICAL SPECIFICATIONS

Core chip: AMD Zynq UltraScale+ XCZU19EG

Quad-core Cortex-A53 · 522,720 LUT / 1,968 DSP

Image pipeline: Correction → hue mask → box and first moment

Processor software: detection, coordinate mapping, speech

Peripherals: camera / audio, robotic arm, gigabit Ethernet

Tools: Vivado 2020.1, Ubuntu 22.04, ROS 2, PyTorch

RESULTS

✓ VERIFIED ON HARDWARE — SORTING SUCCEEDED

7.26 fps mean publish rate

≈187 ms capture to publish

Colour chain only · 1280×720 in, 640×360 processed

661 assertions passed · gate module 866 LUT / 949 FF / 0 BRAM / 0 DSP

7,292 LUT integrated build · WNS +0.632 ns · routed

RESULTS

OUTLOOKS

1. Four-function build onto the board, tested on real samples
2. Direct data path for higher pixel throughput

OUTLOOKS

AMD